Chapter 2
SOFTWARE REQUIREMENTS SPECIFICATION

The Software Requirements Specification (SRS) is the document that states, in verifiable terms, what MezFlow must do and how well it must do it. It turns the problem described in Chapter 1, namely ordering that is fragmented across diners, limited kitchen visibility, digital tools that stop at the menu or the kitchen ticket, and table turnover that is assumed and not tracked, into a set of numbered functional and non-functional requirements that the team can design, implement and test against [11].

This SRS is important for the whole software development life cycle. It gives the three team members, the supervisor and the evaluators one agreed description of the system boundary, reduces the risk of misunderstanding between developers and stakeholders, and supplies the basis for the functional models in Chapter 4, the design in Chapter 5 and the test cases in Chapter 7. Every requirement is written so that it can be checked by a test, an inspection or a demonstration.

2.1 Introduction

The SRS forms a foundation of the MezFlow project. It specifies the features, operating conditions and performance targets of a real-time restaurant dining and kitchen operations management system for small and medium dine-in restaurants. The system connects a zero-install customer Progressive Web App (PWA), a single-screen Kitchen Display System (KDS), a management console and a backend REST and WebSocket service, and it follows the workflow of Customer, Table Session, Order, Kitchen, Preparation, Serving, Bill, Payment, Table Cleanup and Available.

The purpose of this chapter is to give every participant a clear and consistent view of what has to be built. The requirements are derived from the project proposal, from the objectives, problem statement and proposed solution in Chapter 1, and from the review of ten existing restaurant technology products. They are written in the form “the system shall …”. Functional requirements carry the identifier FR-Mx-nn, where Mx is the module number (M1 to M12) and nn is the sequence number inside that module, and non-functional requirements carry the identifier NFR-nn. Each requirement describes one capability or one measurable quality, so that it can be traced to a design element and to a test case.

2.1.1 Document Scope

The scope of this document is to lay out a complete framework of requirements for MezFlow, so that developers, the supervisor and other stakeholders share the same understanding of its purpose, behaviour and quality targets. The document covers the five user classes (customer, kitchen staff, service staff, cashier and manager) and what each of them may do; the functional requirements of the twelve modules of the system, which are Identity and Access Control, Digital Menu and QR Table Access, Collaborative Table Ordering, Kitchen Display System, Order Lifecycle and Real-Time Synchronization, Preparation-Time Tracking and Delay Detection, AI-Assisted Dish Explanation, Billing and Payment Workflow, Table Lifecycle Management, Management Console, Service Requests and Notifications, and Analytics and Operational Records; the non-functional requirements for performance, reliability, security, usability, maintainability and compatibility; the legal, platform and deployment constraints; and the techniques and the time frame of the requirement-gathering phase.

The pilot deployment targets dine-in service in independent, small and medium restaurants, cafés and food-court outlets with roughly 5 to 30 tables. As stated in Chapter 1, the following items are outside the scope of this SRS: food delivery and rider marketplace integration, accounting, Enterprise Resource Planning (ERP) and payroll systems, inventory and supply-chain management beyond basic availability toggles, custom POS hardware, biometric or facial-recognition identification, native mobile applications, and automated detection of physical table cleanliness. Payment is recorded by a cashier as cash or card, and an online payment gateway is not part of the core scope. The requirements are delivered in two phases: FYP-I covers the foundation and the ordering path (Modules 1 to 4 and the groundwork of Module 8) and FYP-II completes Module 8 and adds Modules 5 to 7 and 9 to 12.

2.1.2 Audience

This document is written for several groups of readers. The **development team** (Sheikh Muhammad Khizar, Talha Mumtaz and Syed Sail Haider Gardezi) uses it as the master plan for design, implementation and integration. The **project supervisor**, Muhammad Farooq, and the **FYP committee and evaluators** use it to check that the scope is clear, achievable and measurable. The **testers**, who are the team members themselves and the participants of the usability trial, use it to derive test cases and acceptance criteria. Finally, **restaurant owners**

and managers who may host or evaluate the pilot, and **future maintainers** who extend the platform, can use it to understand what the system promises and where its limits lie.

The system itself is designed for the user classes in Table 2.1. Each class has a role that limits what it can see and do.

Table 2.1: User Classes of MezFlow

User Class	Description	Main Need from the System
Customer	A diner who scans the table QR code on a mobile phone, alone or in a group sharing one table.	Order at their own pace, understand unfamiliar dishes, share one cart, follow the order live, request the bill and call for service.
Kitchen Staff	Cooks and kitchen supervisors who prepare orders at one or more stations (for example Grill, Fry, Hot Kitchen, Beverage).	One organised queue of all active orders, with filters and automatic delay flags instead of scattered paper tickets.
Service Staff	Waiters and floor staff who respond to guests and reset tables.	Fewer repeated trips, a queue of service requests and a live table-status view, with a cleanup confirmation step.
Cashier	Staff member who prepares and settles bills.	An accurate, itemised bill tied to a verified table session and a way to record cash or card payment.
Manager	Restaurant manager or owner’s representative who configures and supervises the restaurant.	One console for the menu, prices, tables, staff and settings, plus analytics and an audit trail.

2.2 Functional Requirements

The functional requirements of MezFlow define the exact operations and capabilities that the system must provide to deliver its purpose. They cover secure access, QR-based menu access, collaborative table ordering, kitchen operations with delay detection, real-time synchronisation of orders and tables, grounded AI dish explanations, billing, table lifecycle management, administration, service requests and analytics.

2.2.1 Functional Requirements

Table 2.2 summarises which user class may use which group of capabilities. Access of the manager to operational functions is meant for oversight and overrides, and every such action is recorded in the audit log (FR-M1-06).

Table 2.2: Capability Access by User Class

Capability	Customer	Kitchen	Service	Cashier	Manager
Scan the table QR code, join the table session, build the shared cart and submit the order	Yes	–	–	–	–

Capability	Customer	Kitchen	Service	Cashier	Manager
Use “Explain This Dish”, follow live order status, request the bill and send service requests	Yes	–	–	–	–
View active orders on the KDS and advance their status	–	Yes	–	–	Yes
Respond to service requests, view live table status and confirm table cleanup	–	–	Yes	–	Yes
Review bill requests, record cash or card payment and close the session	–	–	–	Yes	Yes
Manage menu, prices, preparation times, tables, QR codes, staff and restaurant settings	–	–	–	–	Yes
View analytics, the audit trail and exportable reports	–	–	–	–	Yes
Manage own profile (preferred language and notification preferences)	Yes	Yes	Yes	Yes	Yes

The requirements of each module are listed below. A module is marked with the phase (FYP-I or FYP-II) in which it is delivered.

1. Identity and Access Control (M1, FYP-I):

This module controls who may enter the system and what each person may do. It provides sign-up and login with one-time password (OTP) verification, JSON Web Token (JWT) sessions, five roles and an audit log of staff actions [9].

Table 2.3: Functional Requirements of M1: Identity and Access Control

ID	Requirement
FR-M1-01	The system shall allow a user to sign up and log in using a phone number or an e-mail address, verified by a one-time password.
FR-M1-02	The system shall issue a JWT-based session after a successful login and shall expire it after a defined period of inactivity.
FR-M1-03	The system shall support the five roles Customer, Kitchen Staff, Service Staff, Cashier and Manager, and shall enforce role-based access control on the server for every API request and every WebSocket connection.
FR-M1-04	The system shall let each user set a preferred language (Urdu or English) and notification preferences in a profile.
FR-M1-05	The system shall store any password only as a bcrypt hash and shall never return or log it in plain text.
FR-M1-06	The system shall write an audit-log entry, containing the user, the action and the time, for every staff action such as menu changes, order overrides and bill approvals.
FR-M1-07	The system shall associate every user, table, session and order with one restaurant and shall prevent a user from reading or changing the data of another restaurant.

2. Digital Menu and QR Table Access (M2, FYP-I):

This module lets a diner open the restaurant's menu directly from a table QR code, in the browser and without installing an application [3, 4].

Table 2.4: Functional Requirements of M2: Digital Menu and QR Table Access

ID	Requirement
FR-M2-01	The system shall generate one unique QR code per table that identifies the restaurant and the table and opens the customer PWA in the phone browser.
FR-M2-02	The system shall display menu categories, item descriptions, images, prices and live availability.
FR-M2-03	The system shall support structured modifiers for each item: spice level, quantity, size, add-ons, ingredient removal and a free-text note.
FR-M2-04	The system shall reflect a price or availability change made by the manager in the customer menu without any action by the diner other than normal use of the app.
FR-M2-05	The system shall prevent an unavailable item from being added to a cart.
FR-M2-06	The customer interface shall be a mobile-first PWA that does not require installation from an app store.

3. Collaborative Table Ordering (M3, FYP-I):

This module lets several diners at one table build one live order together while the system still knows who added what.

Table 2.5: Functional Requirements of M3: Collaborative Table Ordering

ID	Requirement
FR-M3-01	The system shall let every diner who scans the same table QR code join the single active session of that table, and shall create the session if none is active.
FR-M3-02	The system shall identify each diner inside a session so that every cart item can be attributed to the diner who added it.
FR-M3-03	The system shall let diners add, change and remove items in one shared cart, and shall record the diner responsible for each action.
FR-M3-04	The system shall show every cart change to all diners at the table, with the server holding the authoritative cart state.
FR-M3-05	The system shall create exactly one order per table submission and shall pass it to the kitchen with the table number, items and modifiers.
FR-M3-06	The system shall reject any request that tries to read or change a session that does not belong to the table the diner scanned.

4. Kitchen Display System (KDS) (M4, FYP-I):

This module replaces paper tickets with a single-screen, software-only queue that runs in an ordinary browser [6, 7].

Table 2.6: Functional Requirements of M4: Kitchen Display System (KDS)

ID	Requirement
FR-M4-01	The system shall show every active order on one screen as an independent card.
FR-M4-02	Each card shall display the order number, table number, items, modifiers, elapsed time and status.
FR-M4-03	The system shall let kitchen staff filter cards by status (New, Preparing, Ready, Delayed).
FR-M4-04	The system shall let kitchen staff filter cards by kitchen category (for example Grill, Fry, Hot Kitchen, Beverage).
FR-M4-05	The system shall let kitchen staff move an order to its next status from its card.
FR-M4-06	The system shall display at least 10 simultaneous table orders submitted within a 2-minute window correctly and without interference between orders.
FR-M4-07	The KDS shall run on any browser-capable tablet, laptop or low-cost computer and shall be available only to Kitchen Staff and Manager users.

5. Order Lifecycle and Real-Time Synchronization (M5, FYP-II):

This module models the order as a state machine and keeps every connected screen in step by means of WebSocket events [8, 28].

Table 2.7: Functional Requirements of M5: Order Lifecycle and Real-Time Synchronization

ID	Requirement
FR-M5-01	The system shall implement the order lifecycle New, Confirmed, Preparing, Ready and Dispatched, and shall reject any transition that is not allowed.
FR-M5-02	The system shall record every order state change with the user and the time.
FR-M5-03	The system shall push every order, kitchen and table-status change over WebSockets to the customer app, the KDS and the management console within 2 seconds under normal network conditions.
FR-M5-04	The system shall show the diner the live order status without a page refresh.
FR-M5-05	The system shall handle simultaneous orders from different tables independently, with no cross-order interference.
FR-M5-06	The system shall reconnect a dropped client automatically, resynchronise its state with the server and offer a REST fallback for critical reads.
FR-M5-07	The system shall authenticate every WebSocket connection and shall send events only to clients authorised for that restaurant and table.

6. Preparation-Time Tracking and Delay Detection (M6, FYP-II):

This module measures how long the kitchen takes and flags orders that exceed the expected time.

Table 2.8: Functional Requirements of M6: Preparation-Time Tracking and Delay Detection

ID	Requirement
FR-M6-01	The system shall let the manager configure an expected preparation time for every menu item.

ID	Requirement
FR-M6-02	The system shall calculate elapsed time automatically from the moment preparation starts.
FR-M6-03	The system shall flag an order as Delayed on the KDS when its elapsed time exceeds the expected time of its items.
FR-M6-04	The system shall map a customer-facing category (for example Main Course) to an internal kitchen station (for example Hot Kitchen) so that items are routed to the correct station.
FR-M6-05	The system shall store delay information so that it can be used in analytics (FR-M12-01).

7. AI-Assisted Dish Explanation (M7, FYP-II):

This module explains a dish to the diner in natural language while staying strictly inside the information that the restaurant has supplied [10].

Table 2.9: Functional Requirements of M7: AI-Assisted Dish Explanation

ID	Requirement
FR-M7-01	The system shall provide an “Explain This Dish” action on every menu item that has explanation data.
FR-M7-02	The system shall generate the explanation only from the restaurant-supplied fields: ingredients, flavour, texture, heat level and portion size.
FR-M7-03	The system shall produce the explanation in English or Urdu, as chosen by the diner.
FR-M7-04	The system shall let the manager choose which fields are supplied to the AI feature for each dish.
FR-M7-05	The system shall not state a fact that is absent from the supplied fields and shall say that the information is not available when a field is missing.
FR-M7-06	The system shall allow a manager to review a dish explanation before it is published to customers.
FR-M7-07	The system shall keep the menu usable and show the standard item description if the AI service is unavailable.

8. Billing and Payment Workflow (M8, FYP-I and FYP-II):

This module takes a table session from the request for the bill to its closure. In FYP-I the data model for bills and payments and the server-side price validation are built; in FYP-II the bill request, cashier payment and session closure flow is completed.

Table 2.10: Functional Requirements of M8: Billing and Payment Workflow

ID	Requirement
FR-M8-01	The system shall let a diner request the bill from the customer app.
FR-M8-02	The system shall recompute every price, tax and service charge on the server from the menu and restaurant settings and shall never trust a price sent by the customer device.
FR-M8-03	The system shall show the cashier the bill requests with an itemised bill that is tied to the verified table session.
FR-M8-04	The system shall let an authorised cashier record payment as cash or card.

ID	Requirement
FR-M8-05	The system shall close the digital session after payment and shall keep a full audit trail of the bill, the payment and the closure.
FR-M8-06	The system shall not mark the table as Available when the session is closed; it shall move the table to Awaiting Cleanup (FR-M9-03).

9. Table Lifecycle Management (M9, FYP-II):

This module tracks the physical table independently from the digital session, so that payment alone never frees a table.

Table 2.11: Functional Requirements of M9: Table Lifecycle Management

ID	Requirement
FR-M9-01	The system shall implement the table lifecycle Available, Occupied, Awaiting Cleanup and Available as a state machine that is independent of the order lifecycle.
FR-M9-02	The system shall set a table to Occupied when a table session starts.
FR-M9-03	The system shall set a table to Awaiting Cleanup when the session is closed after payment.
FR-M9-04	The system shall return a table to Available only after a Service Staff or Manager user confirms that it has been cleaned.
FR-M9-05	The system shall provide service staff and managers with a live table-status view.
FR-M9-06	The system shall record every table state change with the user and the time.

10. Management Console (M10, FYP-II):

This module is the single administration interface of the restaurant, available to the manager role only.

Table 2.12: Functional Requirements of M10: Management Console

ID	Requirement
FR-M10-01	The system shall let the manager create, edit and remove menu items, categories, prices, availability and preparation times.
FR-M10-02	The system shall let the manager create tables, generate and print their QR codes and monitor active table sessions.
FR-M10-03	The system shall let the manager create staff accounts, assign one of the staff roles and deactivate an account.
FR-M10-04	The system shall let the manager configure restaurant-level settings: taxes, service charges and operating hours.
FR-M10-05	The system shall restrict the console to Manager users and shall show a manager only the data of their own restaurant.

11. Service Requests and Notifications (M11, FYP-II):

This module lets diners ask for help from the table and keeps them informed [24].

Table 2.13: Functional Requirements of M11: Service Requests and Notifications

ID	Requirement
FR-M11-01	The system shall let a diner send a service request for a waiter, a water refill or other assistance.
FR-M11-02	The system shall send push notifications for order status, bill readiness and acknowledgement of a service request, according to the notification preferences of the user.
FR-M11-03	The system shall show service staff a notification queue sorted by table and waiting time.
FR-M11-04	The system shall let service staff acknowledge and complete a service request and shall show the new state to the diner.

12. Analytics and Operational Records (M12, FYP-II):

This module gives management a view of how the restaurant performs and keeps a reliable operational record.

Table 2.14: Functional Requirements of M12: Analytics and Operational Records

ID	Requirement
FR-M12-01	The system shall provide a dashboard of the number of orders, the average preparation time and the delayed-order rate by date and by category.
FR-M12-02	The system shall keep an operational audit trail of order creation, state changes, payments and table status changes.
FR-M12-03	The system shall let the manager export reports for management review in a common file format such as CSV.
FR-M12-04	The system shall treat audit records as append-only, so that a user cannot edit or delete them through the application.

2.2.2 Traceability to Project Objectives

Table 2.15 links the objectives stated in Section 1.3 to the modules and the non-functional requirements that satisfy them, so that no objective is left without a requirement. Modules 11 and 12 support the operation of the restaurant and the evaluation of the system and are not tied to a single objective.

Table 2.15: Traceability between Objectives and Requirements

Project Objective	Modules	Related NFRs
Zero-install, QR-activated PWA with a first-load time under 3 seconds	M2	NFR-01
Collaborative table sessions with per-diner attribution of items	M3	NFR-06
Real-time propagation of order, kitchen and table changes within 2 seconds	M5, M3, M4	NFR-02, NFR-05
Single-screen KDS with category and status filtering for at least 10 simultaneous orders	M4	NFR-04
Preparation-time tracking and automatic delay flags	M6	NFR-02

Project Objective	Modules	Related NFRs
Independent, auditable order and table state machines	M5, M9	NFR-06, NFR-10
Digital billing workflow that does not assume the table is free	M8, M9	NFR-09
Grounded “Explain This Dish” feature in English and Urdu	M7	NFR-12, NFR-13
Management console protected by role-based access control	M10, M1	NFR-07, NFR-08
Evaluation by load testing (50 WebSocket connections) and a usability trial (SUS of at least 70)	All modules	NFR-03, NFR-11

2.3 Non-Functional Requirements

2.3.1 Software Quality Attributes

The non-functional requirements describe how well MezFlow must perform its functions. Where the project proposal gives a measurable target, the same figure is used here. Table 2.16 lists the quality attributes together with the way in which each is verified.

Table 2.16: Non-Functional Requirements

ID	Attribute	Requirement	Verification
NFR-01	Performance (page load)	The customer PWA shall complete its first load in under 3 seconds on a typical mobile connection.	Browser performance audit on a throttled mobile profile.
NFR-02	Performance (real time)	Order, kitchen and table-status changes shall reach every connected client within 2 seconds under normal network conditions [8].	Timestamp comparison during load tests.
NFR-03	Concurrency (connections)	The system shall serve at least 50 simultaneous WebSocket connections without state corruption.	Locust load test.
NFR-04	Concurrency (orders)	The system shall process at least 10 simultaneous table orders submitted within a 2-minute window with no cross-order interference.	Load scenario and inspection of the KDS.
NFR-05	Reliability	After a connection drop the client shall reconnect automatically and resynchronise its state, and critical reads shall have a REST fallback.	Fault test that disconnects a client during an active order.
NFR-06	Data integrity	The server shall be the single source of truth for cart, order, table and price data, and every state transition shall be validated and applied atomically.	Unit and integration tests of the state machines and price validation.
NFR-07	Security (access)	The system shall use JWT sessions, bcrypt password hashing and server-side role-based access control for every API and WebSocket endpoint [9, 29].	API tests for every role.
NFR-08	Security (tenant isolation)	The data of each restaurant shall be isolated by a tenant identifier enforced with PostgreSQL row-level security, so that no user can reach another restaurant’s data.	Integration test with two restaurants.

ID	Attribute	Requirement	Verification
NFR-09	Security (transport and price)	All traffic shall use HTTPS through the Nginx proxy, and the backend shall never trust a price sent by a client.	Configuration review and price-validation tests.
NFR-10	Auditability	Every staff action and every order, table and payment state change shall be logged with actor and time, and the log shall be append-only.	Audit-log inspection tests.
NFR-11	Usability	The system shall reach a System Usability Scale score of at least 70 from at least 10 participants, including non-technical users, and kitchen and service screens shall need only brief on-site training [14].	SUS usability trial.
NFR-12	Localisation	The customer interface and the AI explanations shall be available in English and Urdu, and Urdu text shall display correctly from right to left.	Manual review of both languages.
NFR-13	AI grounding	A dish explanation shall contain no fact that is absent from the restaurant-supplied data [10].	Manual review of generated explanations.
NFR-14	Maintainability	The system shall use a modular layered architecture, versioned database migrations and automated unit, API and integration tests that run on every code change.	CI pipeline results (GitHub Actions, pytest).
NFR-15	Compatibility and portability	The customer PWA, the KDS and the console shall run in the current versions of mainstream browsers on phones, tablets and laptops, and the backend shall be deployable with Docker.	Cross-browser checks and container deployment.
NFR-16	Scalability	The pilot shall support restaurants with roughly 5 to 30 tables, and the multi-tenant design shall allow more restaurants to be added without a redesign.	Design review and load tests.
NFR-17	Availability	The system shall be available throughout the operating hours configured by the restaurant, and planned maintenance shall take place outside those hours.	Deployment checks and monitoring during the pilot.
NFR-18	Information clarity	The interface shall show clear messages for network loss, unavailable items, invalid actions and rejected state transitions.	Manual test of error scenarios.

2.3.2 Other Non-Functional Requirements (e.g. Legal, Platform etc.)

Besides the visible features, the trustworthiness of a restaurant platform depends on legal, platform and deployment conditions. The following requirements and constraints apply to MezFlow.

- **Legal and privacy:** The system shall collect only the personal data that it needs, which is a phone number or an e-mail address and a display name, and shall protect it according to the data protection requirements that apply where the restaurant operates. The system shall not store card numbers, because card payment is recorded by the

cashier and is not processed online. The use of third-party libraries shall respect the licences of the open-source components.

- **Platform:** The customer experience is delivered as a PWA and not as a native application [4]. The KDS and the management console are web applications that run on a tablet, laptop or low-cost computer that the restaurant already owns, so no proprietary hardware is required.
- **Technology:** The system shall be built with React/Next.js, TypeScript and Tailwind CSS for the interfaces, Python and FastAPI for the REST and WebSocket backend, and PostgreSQL for the data [26, 30, 31].
- **Deployment:** The system shall be packaged with Docker and Docker Compose and served behind Nginx on a cloud virtual private server, with continuous integration through GitHub Actions.
- **Connectivity:** A working internet connection is required. Unstable restaurant Wi-Fi is handled by automatic reconnection and state resynchronisation and not by full offline operation [24].
- **Language and localisation:** The interfaces shall support English and Urdu, and monetary amounts, taxes and service charges shall follow the configuration of each restaurant.
- **Personalisation:** Each user shall be able to set a preferred language and notification preferences in the profile.

Together, these conditions keep the system lawful, affordable to self-host, easy to start using and secure against unauthorised access and unintended data exposure.

2.4 Requirement Gathering Techniques Used

The requirements of MezFlow are gathered with several complementary techniques so that the result is complete and accurate. A review of existing products shows what the market already offers, interviews with restaurant staff show how the work is done in practice, usability walkthroughs with prospective diners show how customers expect to order, and regular reviews with the supervisor confirm and refine the outcome. Every finding is written as a user story in the product backlog of the Agile Scrum process described in Section 1.8 [13].

2.4.1 Review of Existing Systems

Ten widely used restaurant technology products were reviewed: Toast, Square for Restaurants, mr yum, Lightspeed Restaurant, TouchBistro, Revel Systems, Epos Now, Flipdish, Menu Tiger and Oracle MICROS Simphony. Only the features that the vendors document in their public pages were compared. This review, reported in Section 1.2.1, showed which functions are well covered (QR ordering, POS and kitchen display) and which are not documented in the reviewed sources: a live shared cart with per-diner attribution, a software-only kitchen display that runs on any browser, live order status for the diner, a table tracked until it is cleaned after payment, and grounded AI dish explanations. These gaps were turned into the requirements of Modules 3, 4, 5, 7 and 9. Industry survey data [12] was used to confirm the demand for kitchen technology and QR menus.

2.4.2 Interviews with Restaurant Staff

Informal, semi-structured interviews with restaurant staff serve as the main technique for understanding the real workflow. The interviewees are drawn from each user class: owners and managers, kitchen staff, waiters, and cashiers. Open-ended questions encourage the interviewees to describe their work freely, and the answers are analysed to find common needs and pain points. The interviews cover the following themes:

- how orders are taken at present and how they reach the kitchen, including how modifiers such as spice level and removed ingredients are communicated;
- how the kitchen keeps track of orders during a rush and how a late dish is noticed;
- how customers ask about unfamiliar dishes, and which language (Urdu or English) they prefer;
- how bills are prepared and paid, and how a table is cleaned and made ready for the next guests;
- which devices and internet connection the restaurant already has, and how much training staff can accept.

The interviews with restaurant staff in Pakistan also check whether the market figures cited in Chapter 1, which come from United States surveys, hold for local demand [12].

2.4.3 Usability Walkthroughs with Prospective Diners

Short walkthroughs with prospective diners test how people expect to scan a QR code, browse the menu, share a cart and follow an order. Low-fidelity screen designs prepared in Figma are shown to the participants, who are asked to complete simple tasks while the team

observes where they hesitate. The observations shape the requirements of Modules 2, 3, 7 and 11 and the usability target in NFR-11.

2.4.4 Supervisor Reviews and Document Analysis

The project proposal and Chapter 1 were analysed as the first written statement of the requirements. Every sprint ends with a sprint review in which the supervisor checks the working increment and the updated requirements, and the outcome of the review is fed back into the product backlog [13].

2.5 Time Frame

The requirement phase is planned to take about four weeks in total, because some phases run in parallel, and it ends with the SRS document that is presented at the mid-phase review. Table 2.17 shows the duration of each phase. With the requirements gathered and understood, the team can move on to functional modelling, design and implementation of the first three modules, which together make up about 25% of the functionality.

Table 2.17: Time Frame of the Requirement Phase

Sr. No	Phase	Duration
1	Project requirement initiation (scope taken from the project proposal)	3 Days
2	Review of existing systems	5 Days
3	Interviews with restaurant staff	5 Days
4	Usability walkthroughs with prospective diners	3 Days
5	Requirement analysis (modules turned into numbered requirements)	5 Days
6	Requirement validation with the supervisor	3 Days
7	SRS document	5 Days

MezFlow is designed with real restaurants in mind. Its requirements ask for a system that is fast enough for a busy dining room, secure enough to protect the data of every restaurant, simple enough for staff who receive only brief training, and open enough to run on the devices that a restaurant already owns. The team remains committed to delivering a product that meets all functional and non-functional requirements stated in this chapter.